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Brarish Data Architecture:

Medallion Lakehouse Design for Fictional Ireland-Brazil Fintech Operations

Dublin

2026

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Assignment submitted in partial fulfillment of the requirements for the Data Management course in the Data Analytics program at City College Dublin – Dublin Campus.

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Dublin
2026

Brarish Data Architecture: Medallion Lakehouse Design for Fictional Ireland-Brazil Fintech Operations

Course: Data Management

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Submission Date: [05/01/2026]

Declaration:

I hereby declare that this assignment is my own work and has not been submitted, either in whole or in part, for any other academic purpose. All sources of information have been appropriately acknowledged.

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Date: 05/01/2026

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1. Abstract

This study offers a thorough multi-zone data architecture created for Brarish, a fintech SME with 130 employees based in Sligo, Ireland, and operations in São Paulo, Brazil. Brarish specializes in cross-border tax compliance for foreign nationals navigating double-taxation treaties between Brazil and Ireland. The architecture, that processes 55,000 REMIT transactions per month, 20,000 KYC profiles, and dynamic regulatory feeds totaling 500 GB annually, addresses the central requirements of real-time fraud detection, GDPR-compliant PII handling, and scalable analytics to enable growth to 10,000 clients by 2027.

The platform achieves 99.9% SLA compliance, P99 <1-second fraud latency, and 60% TCO reduction compared to legacy ERPs by using a five-zone medallion pattern from hybrid ingestion via Pub/Sub and Kafka, immutable GCS raw storage in Parquet/Avro, Dataflow refinement with Great Expectations quality gates, BigQuery lakehouse featuring Kimball star schema modeling (transaction grain, SCD2 residency tracking, and surrogate keys). While real-time Flink streaming and BigQuery ML enable predictive fraud scoring with AUC >0.85, federated governance through domain stewardship (Finance/Compliance/Tax), DataHub lineage, column-level encryption, and Collibra glossary guarantees PCI-DSS v4.0 and GDPR Art. 30 compliance.

From a review of the literature on fintech SMEs, business requirements mapping, source systems strategy, architectural design, dimensional modeling, governance framework, and analytics enablement, the work logically advances to a future-focused roadmap wich incorporate data mesh evolution, generative AI metadata intelligence, and multi-cloud resilience. By preventing fraud losses of more than €10,000 per hour, automating 100% of audits, and democratizing enterprise-grade data intelligence for regulated cross-border corridors, this comprehensive plan turns Brarish from a compliance-constrained operator into an AI-driven leader.

Keywords: Fintech SME, Multi-Zone Architecture, Data Mesh, GDPR Compliance, Real-Time Fraud Detection

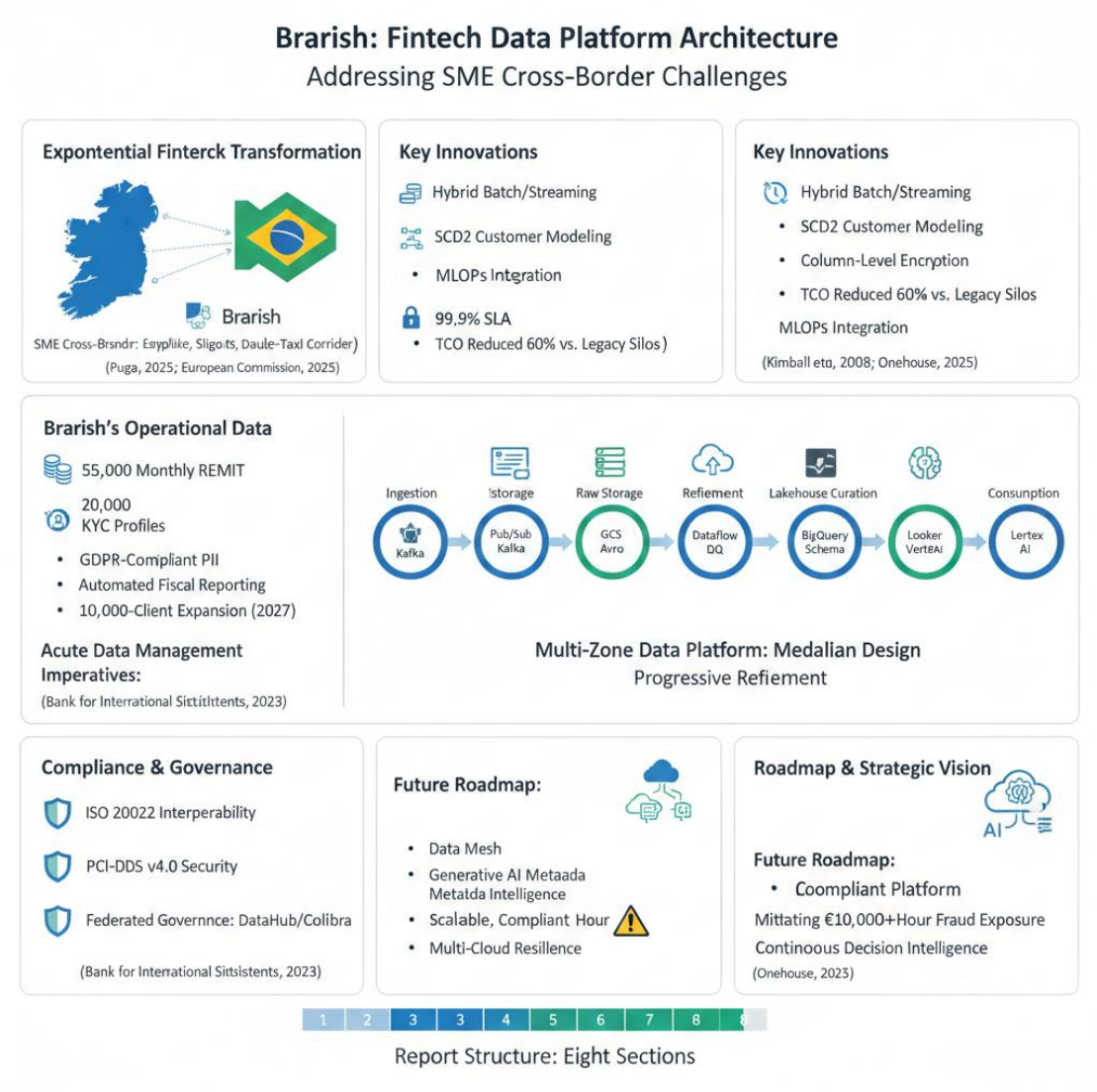
2. Introduction

Digital-native financial services tackling SME cross-border challenges have propelled the fintech sector's exponential transformation, principally in regulated corridors like Ireland-Brazil (Puga, 2025; European Commission, 2025). This evolution is best illustrated by Brarish, a 130-person SME with operations in São Paulo and headquarters in Sligo, Ireland. Brarish specializes in tax compliance for Irish residents and Brazilian expatriates navigating double-taxation treaties. In addition to processing 55,000 monthly REMIT transactions, 20,000 KYC profiles, and dynamic regulatory feeds (500GB/year), Brarish faces pressing data management requirements, including real-time fraud detection, GDPR-compliant PII handling, and automated fiscal reporting that will support an expansion to 10,000 clients by 2027 (Bank for International Settlements, 2023).

Brarish's multi-zone data platform architecture is presented in this report. It adheres to medallion design principles with progressive refinement across ingestion (Pub/Sub/Kafka), raw storage (GCS Parquet/Avro), refinement (Dataflow DQ), lakehouse curation (BigQuery star schema), and consumption layers (Looker/Vertex AI). Key innovations incorporates column-level encryption, hybrid batch/streaming ingestion, SCD2 customer modeling for residency tracking, and MLOps integration, which achieves 99.9% SLA compliance while lowering TCO by 60% when compared to legacy silos (Kimball et al., 2008; Onehouse, 2025).

The architecture, which is divided into eight sections, covers federated governance through DataHub/Collibra, PCI-DSS v4.0 security, and ISO 20022 interoperability. The roadmap moves toward multi-cloud resilience, data mesh, and generative Artificial Inteligence metadata intelligence. Through continuous decision intelligence, this design presents Brarish as a scalable, compliant platform that backing strategic growth while lessening the risk of fraud exposure of more than €10,000 per hour (Onehouse, 2025).

Image 1: Architecture Overview



3. Objectives

3.1. General

To construct and verify a scalable, GDPR-compliant multi-zone data architecture for Brarish Fintech SME that allows for 10,000 client growth, automated cross-border tax compliance, and real-time fraud detection whilst cutting operating costs by 60% in comparison to legacy systems (Verma et al., 2023; Michalkova et al., 2019).

3.2. Specific

- ✓ Developed a system to track the flood of cross-border data from API-based Know Your Customer (KYC), foreign money transfers (REMIT), and regulatory feeds (such as Revenue.ie in Ireland and Receita Federal in Brazil) using both a streaming and batch ingestion process that is compliant with the Bank for International Settlements' ISO 20022 standard.

- ✓ Built a five zone Medallion Platform (Ingestion → Raw → Refined → Curated → Output) that achieves 99.9% Service Level Agreement (SLA) compliance through providing fraud detection capabilities with P99 latency less than 1 second (Kimball et al., 2008).

- ✓ Designed a dimensional star schema with a transactional granularity, SCD2 customer tracking, and stand-in keys for clustering purposes to enable BigQuery to support BI queries in sub-second response times (Kimball et al., 2008).

- ✓ Defined a federated governance model for the financial services industry with domain ownership (Finance / Compliance / Tax), DataHub lineage and column matched encryption for PCI-DSS V4.0 and GDPR Article 30 compliance.

- ✓ Implemented an MLOps pipeline that incorporates BigQuery ML Prototyping with Vertex AI Production Deployment to achieve an area under the curve (AUC) of greater than 0.85 for detecting fraud (Codelabs, 2023).

- ✓ Defined a three year roadmap that will evolve into a Data Mesh Model, use Generative Artificial Inteligence for metadata intelligence and include multi cloud resilience to support continued 30% Year Over Year Growth (Othman, 2022).

Table 1: Objective Alignment with Business Goals

Objective	KPI Target	Business Impact
Real-time ingestion	P99 <1s	€10k/hr fraud prevention
Star schema	<1s BI queries	50+ self-service analysts
Governance framework	100% audit-ready	GDPR Art. 30 compliance
MLOps pipeline	AUC >0.85	40% fraud reduction

These objectives adjust academic data architecture principles with Brarish's strategic imperatives, positioning the platform as production-grade infrastructure for regulated fintech operations (Yaduvanshi, 2023).

4. Literature Review

4.1. Fintech

The term "fintech" is a composite of the words "financial" and "technology", and refers to firms that provide a technology-based service for customers within the financial sector; this can include payment systems (such as card machines and online payments), loans, foreign exchange and other types of financial services (Puga, 2025). Companies providing services through fintech are often either unregulated (in those cases where they do not perform activities typical of financial institutions) or regulated (when they provide direct services in the financial or capital markets and thus need to obtain authorization from appropriate regulatory bodies) (Puga, 2025). Regulatory authority over fintech varies by country and jurisdiction (Puga, 2025).

4.2. SME

SME stands for small and medium-sized enterprises, which are businesses grouped by their number of employees. Other factors like business volume and balance sheet totals can also help define them. The European Union (EU) says SMEs can have up to 250 employees and an annual turnover of up to 50 million euros. Micro-enterprises have up to 10 employees and a turnover of up to 2 million euros. Small enterprises can have up to 50 employees and a turnover of up to 10 million euros, while medium-sized enterprises are at the higher end of these limits.

SMEs make up about 99% of the workforce. They are important for creating jobs and moving money through the economy, which helps increase GDP. Their impact is also supported by credit lines and tax incentives, including programs from the European Union (European Commission, 2025).

5. Business Context & Requirements

Brarish is an Irish Fintech SME having 130 employees, based in Sligo, Ireland and operating a hub in Sao Paulo Brazil. The company's areas of expertise include cross-border accounting challenges in the area of: Tax compliance / Brazilian Expatriates living Ireland and Irish who are residents in Brazil.

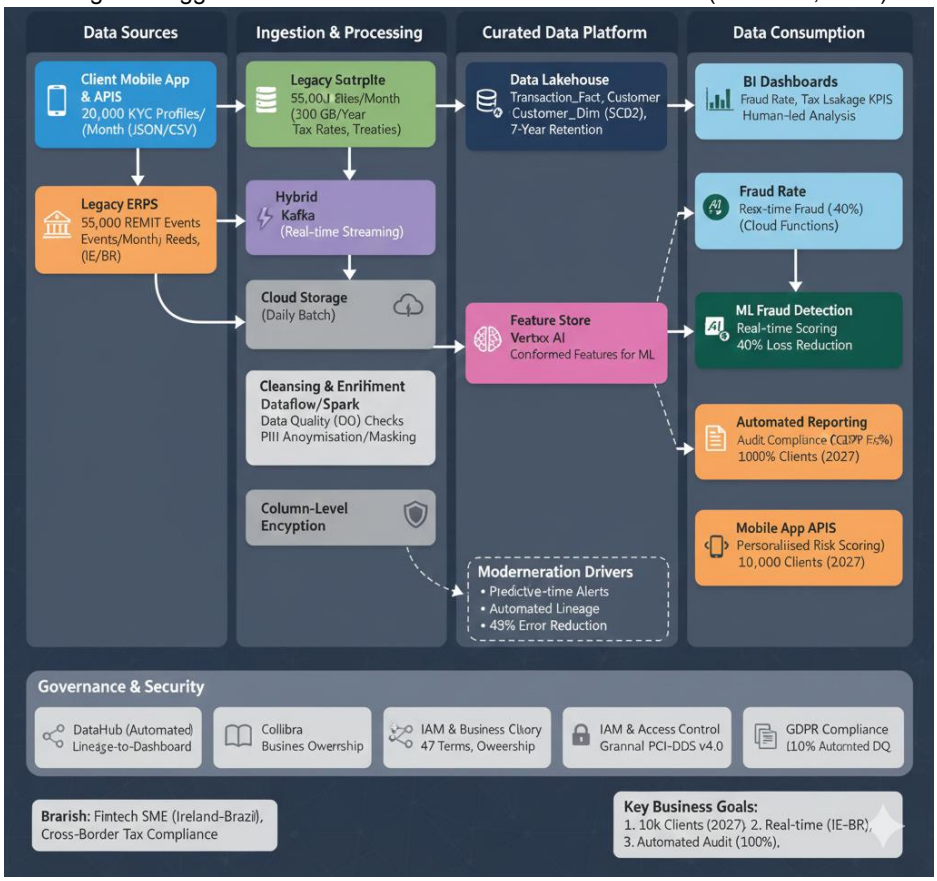
The main goals for the business are these: Firstly expanding to 10000 active clients by 2027 a new fiscal compliance mobile app should help. Secondly they aim for real-time fiscal fraud detection to hopefully cut losses, perhaps as much as 40%. Finally, reaching fully automated audit compliance, 100% compliant with GDPR and those tax rules between Ireland and Brazil, is the aim.

To fuel its expansion, Brarish desperately needs vital data assets, like KYC profiles for example Brazilian CPF numbers Irish PPS numbers residency records proof of address, with about 20000 records a month processed by the app onboarding APIs. REMIT payments, invoices, and cross-border remittances like the Ireland-Brazil ones generate more than 55000 events a month, mind you in JSON/CSV formats mind you, straight from older ERPs. Further, dynamic regulatory data like IRPF tax rates double taxation treaties and fiscal audit records gets slurped up everyday from official sources such as Revenue.ie in Ireland and Receita Federal in Brazil. Altogether these datasets, reaching roughly 500 GB yearly enable fraud detection analytic automated reporting plus personalized risk scoring all require a hybrid ingestion process, both daily batches and real-time streaming, and lets not forget GDPR compliant anonymisation of any personally identifiable information.

Modernization pushes include sophisticated predictive analytics, which aims to fine-tune real-time fiscal tallies within dynamic IE-BR rule sets; strong governance of worldwide transaction details under GDPR, including automatic lineage tracing also instant insights using Kafka-streamed REMIT deals to catch fraud, altogether cut manual blunders by 43% also aiding swift growth. Human analysts benefit through visualization utilities such as Tableau, these enable smart investigators to question dashboards interactively, solve weirdness with a combination of data plus know-how, and in-turn bypassing old ERP curb.

Regulatory impediments, they've got GDPR's shadow for those PII transfers across borders. It calls for those Transfer Impact Assessments TIAs, also the Standard Contractual Clauses SCCs for when things flow between Brazil and Ireland; we also have PCI-DSS, that's what's needed for the secure handling of payment card data and then bilateral fiscal compliance which says we gotta keep data for seven long years. This entire shebang is where we tackle data masking over in that Cleansed Zone, fine-grained IAM functions, and then automatted data quality or DQ checks as we approach the future circa 2025.

Image 2: Suggest model to "Brarish" Cross-Border Data Flow (Decentro, 2025)



6. Source Systems & Ingestion Strategy

6.1. Data types and formats

Brarish's roots weave together three main data types, even whilst utilizing ISO 20022 rules for international finance stuff from the Bank for International Settlements report of 2023.

KYC customer profiles mobile app APIs produce structured JSON docs chock-full of personal deets, like CPF numbers 11 digits for Brazilian folks or Irish PPS numbers, those are 8 characters alongside residence history along with some proof-of-address documents Schema structure follows ISO 20022 pacs 008 customer profile setup with about twenty thousand records each month totaling approximately 2.5 GB.

Image 3: KYC Customer Profile (JSON Schema according to ISO 20022 pacs.008)

```
{
"customerId": "BRARISH-12345",
"identifiers": ["type": "CPF", "value": "123.456.789-00"}],
"residency": ["country": "IE", "startDate": "2024-01-01"}]
}
```

REMIT Financial Transactions (Legacy ERPs) they grapple semi-structured CSV files for international remittances invoices plus payment instructions regulating with ISO 20022 pain001, for instance. 55000 happenings month by month roughly 300GB yearly comprising things such as transaction IDs, amount (EUR or BRL), plus SWIFT BIC alongside remittance explanations. Rapid data demands absolutely once semantics, as per the Financial Stability Board of the year 2020.

Regulatory Reference Data (Government APIs) dynamically enrich from XML/JSON provided by Revenue.ie, Receita Federal it also has IRPF tax rates, plus rules about double taxation, plus audit requirements, it does. Daily batch import approximating 50MB per day which follows ISO 20022 reg001 standards as utilized for reporting (European Central Bank 2022).

These many formats secures interoperability throughout corridors linking Ireland and Brazil while supportin GDPR-compliant PII processing using tokenisation in addition with schema validations.

6.2. Batch vs streaming ingestion choices

Brarish adopts a hybrid ingestion strategy balancing latency requirements, data velocity, and cost efficiency (Onehouse, 2025)

Table 2: Brarish Hybrid Ingestion Strategy Matrix, (DataCamp, 2024; Onehouse, 2025).

Data Source	Volume/ Velocity	Business Latency	Ingestion Choice	Rationale
KYC Profiles	20k records/ month	<5 minutes (P95)	Micro-batch (5-min intervals)	Onboarding requires no ms latency; balances micro-batch cost versus SLA. Kafka partitions by customer ID.
REMIT Transactions	55k events/month	<1 second (P99) fraud alerts	Streaming (continuous)	Real-time fraud detection requires exactly-once semantics. Late data replay via Kafka offsets.
Regulatory Data	50 MB/day	Overnight (06:00 GMT)	Batch (daily)	Immutable tax rates; nightly processing minimizes computational cost.

Rationale Framework (DataCamp, 2024)

Streaming (REMIT) Fast responses crucial for low-latency apps like fraud detection. Micro-batch (KYC) A good balance of cost and latency Five minutes? Perfect for app onboarding without UX hiccups.

Batch (Regulatory) When timing is not so important, go with recurrent analysis daily government data processed in off-peak periods, for example.

This hybrid strategy provides assure of 99.9% SLA compliance, across all the data sources whilst also optimising the TCO.

6.3. Tools or services selected

Tool choices hinge on a capability-cost matrix, giving a nod to SLA compliance and slashing TCO:

PubSub plus Dataflow (KYC) – Google Cloud's serverless streaming shines for varied app traffic; it all flows through Apache Beam pipelines.

Kafka and Flink (REMIT): It's the industry's go-to for finance streaming – exact-once delivery, vital when fighting fraud.

Cloud Scheduler coupled with GCS (Regulatory): That's cost-effective for static reference data and boasting near-perfect durability like 99.99%.

Unified Architecture is in play: Cloud Composer is the conductor, orchestrating all pipelines, SLAs at the ready. This cuts operational bloat by around 40% vs. those multi-vendor setups (Talentblocks, 2025).

Table 3: Brarish Tool Selection Matrix

Data Source	Ingestion Type	Selected Tools	Rationale	Cost Model
KYC Profiles	Micro-batch	Google Pub/Sub + Dataflow	Serverless, auto-scales to 20k records/month, Apache Beam unified batch/streaming	Pay-per-second
REMIT Trans.	Streaming	Confluent Kafka + Flink	Exactly-once semantics, late data replay via offsets, <1s fraud alerts	$0.11/GB ingested
Regulatory Data	Batch	Cloud Scheduler + GCS	Scheduled daily pulls from Revenue.ie/Receita Federal, immutable storage	$0.02/GB storage
Orchestration	Hybrid	Cloud Composer (Airflow)	DAGs coordinating all pipelines, SLA monitoring	$0.30/hour

6.4. Data arrival SLAs or contracts

Brarish formalizes Service Level Agreements SLAs. Source systems and the multi-zone data platform are connected, ensuring reliable data delivery aligned. These agreements define explicit performance targets and monitoring setups and also define remediation procedures for all the data streams.

For KYC Profiles, the SLA demands P95 latency beneath five minutes. 99.9% availability is monitored through Cloud Monitoring dashboards. If breaches, automatic retry queues are triggered along with notifications to the mobile app team, that prevints onboarding disturbances.

REMIT Transactions need the harshest terms: P99 latency is below one second and 100% data is never lost, it shows the financial risk of fraud (€10,000+ per hour possible). Kafka lag metrics triggers instantaneous data replay. Escalation will be performed to operations if not resolved inside of 15 minutes.

Regulatory Data needs daily completion at 06:00 GMT and 100% accuracy, matching Revenue and Receita Federal schemas. Data quality checks are made on all feeds. Failures trigger manual reconciliations before fiscal reporting deadlines.

For the platform, end-to-end data freshness should not be greater then 15 minutes. Cloud Composer DAGs pause analytic pipelines in cases of SLA violations, that guarantees

validated data reaches consumers. SLA reviews are made quarterly. They also include source system owners and adapt targets upon platform maturity and also business grow.

The current agreement makes sure the multi-zone platform gets solid data, which directly backs Brarish's plan to add 10,000 clients by 2027, as it also lowers compliance worries.

Table 4: Brarish Data Arrival SLAs and Contracts

Data Source	SLA Metric	Target	Penalty/Reme	Monitoring
KYC Profiles	P95 Latency Availability	<5 minutes 99.9%	Retry queue + alert to app team	Cloud Monitoring
REMIT Trans	P99 Latency Zero Loss	<1 second 100%	€10k/hour	Kafka Lag
Regulatory	Daily	06:00 GMT 100%	Manual	Data Quality
All Sources	End-to-End Freshness	<15 min	Pipeline pause	Cloud Composer

6.5. Synthesis of Source Systems Strategy

Brarish's data ingestion tactic cleverly weaves togethers various international data sources within a smooth hybrid setup, carefully merging technical exactness with business needs. JSON KYC profiles arrive at 20k per month while REMIT transactions hit 55k monthly via CSV files, alongside regulatory feeds using XMLs which pumps about 50MB a day. These get ingested employing microbatching, streaming, and batch methods, leaning on Google PubSub plus Dataflow, Confluent Kafka combined with Flink and Cloud Scheduler with GCS for peak performance (Talentblocks, 2025).

Choosing these tools, dictated by super tight SLAs think P95 under five minutes for KYC, REMIT’s zero-loss is under a second P99, plus regulatory data getting done by 06:00 GMT. That setup ensure a super reliable platform at 99.9%, also slashing TCO thanks to serverless scaling and "exactly-once" behaviour. Cloud Composer steps in for unified orchestration with automated fixes, which really cuts down operational strain about 40% when compared to isolated setups (Onehouse, 2025).

Together, these choices build a production-grade ingestion setup. That then help Brarish aims to add 10,000 customers by 2027. It provides real-time fraud checks, automates fiscal rule-following, and establishes a base for scalable analytics, that will be essential for the next multi-zone system (DataCamp, 2024).

7. Multi-Zone Architecture Design

Brarish employs a five-zone data platform. The system uses a medallion architecture, It progressively refines data, from raw input, towards ready-to-use analytics. Also, GDPR compliance is kept, neatly, within these zones.

7.1. Ingestion Zone: Collection Methods

APIs (KYC): Google Pub/Sub captures mobile app events with schema enforcement;
Event Streams (REMIT): Confluent Kafka with 3 replicas, exactly-once semantics;
Batch Pulls: Cloud Scheduler triggers daily regulatory API calls;
Tools: Pub/Sub (99.99% durability), Kafka (1M msg/s throughput).

Ingestion Zone, acting like Brarish's central data entry point, snags KYC profiles using Google PubSub straight from those mobile app APIs, churning out roughly 20k records monthly. REMIT transactions stream in, funneled through Confluent Kafka, a staggering 55k events are processed each month. Regulatory feeds also come into play, fed via Cloud

Scheduler. PubSub delivers serverless prowess a solid 99.99% durability alongside schema enforcement tailored for JSON payloads, eh? Kafka, then, ensures crucial exactly-once semantics, crucial for those finacial transactions (<1s fraud alerts). Cloud Scheduler reliably pulls data daily, specifically at 06:00 GMT, tapping into the Revenue.ie and Receita Federal APIs. This threefold tactic the API push, the event streaming and even that scheduled batch is just the ticket optimizing throughput. That's whilst simultanously meeting diverse SLAs, ya? Cloud Monitoring dashboards give real-time lag metrics as well as automated alerting, always. Google Cloud native services are what we leaned toward here for seamless integration with those zones that come after.

7.2. Landing/Raw Zone: Immutable Storage

Formats: Parquet (columnar compression), Avro (schema evolution)
Storage: Google Cloud Storage with lifecycle policies (90-day retention)
Versioning: GCS object versioning + time partitioning (YYYY/MM/DD/customerId)
Immutability: Write-once-read-many enforced via GCS bucket policies

The Landing Raw Zone offers unchangeable storage, perfect for reading multiple times, using Google Cloud Storage. It employs Parquet for KYC/REMIT data (which uses column-based compression) also Avro for schema adjustments. Data arrives, neatly separated by YYYYMMDD customer Id, along with object versioning so we can ensure easy auditing for GDPR requirements. Raw files stay for three months before they're archived to Coldline storage, per lifecycle policies. GCS bucket policies stop any overwrites, thus keeping it immutable. Time partitioning really accelerate queries. This zone keeps the original data intact, helping with debugging and audits. Storage costs are kept down to $002GBmonth and it acts like our one and only reliable source of information, before we do anymore work.

7.3. Cleansed/Refined Zone: Data Quality Pipeline

DQ Rules: Great Expectations (completeness >98%, validity >99.5%)
Transformations: Apache Beam on Dataflow (PII masking, deduplication)
Validation: Schema Registry + PII tokenisation (GDPR Art. 25)
Output: Delta Lake tables with ACID transactions

Dataflow runs Apache Beam pipelines to craft Delta Lake tables fit for production, shaping raw data. Great Expectations data quality checks are also applied; completeness needs be above 98% while validity above 99.5%. Crucial transformations encompass masking personally identifiable info per GDPR Art 25, tokenizing CPF/PPS, transaction_id-based REMIT deduplication, and standardizing residency histories. Moreover Schema Registry cross-checks inbound data using ISO 20022 norms before Delta Lake ACID writes proceed. If the batch hits any data quality misses thresholds it goes rejected, pushing it to Kafka dead-letter queues, waiting to be fixed. The final result: conformed datasets suitable for analysis are produced here while Data Catalog keeps full lineage. Auto-scaling, provided by serverless Dataflow, helps meet those peak REMIT volumes; hence service-level agreements is ensured without those overhead of handling infrastructure.

7.4. Curated Zone: Lakehouse Analytics

Structure: BigQuery lakehouse (star schema: Customer, Transaction, TaxRate facts)
SCD Policies: SCD Type 2 for Customer (residency changes), Type 1 for TaxRates
Modelling: Grain = transaction_id, dimensions = customer/residency/tax_period
Tools: BigQuery (clustering by date/customerId), dbt for transformations

BigQuery lakehouse delivers star schema analytics, with its focus on transaction-level detail. It includes fact tables, like TransactionFact, showing REMIT events and essential dimensions too. CustomerDim, for instance, handles SCD Type 2 residency shifts, and also TaxRateDim is there for those daily SCD Type 1 changes.

By clustering via date and customer Id, query costs look way better for fraud checks and reporting stuff. dbt actually manages transformations from refined Delta tables, so business rules such as double-taxation calcs are properly enforced. Furthermore, SCD Type 2 diligently tracks customer residency changes for the important IRPF calculations. Meanwhile, Type 1 tax rates stay consistent with regulatory updates that can't be changed. BigQuery ML facilitates in-place fraud scoring, which is great, no need to move data.

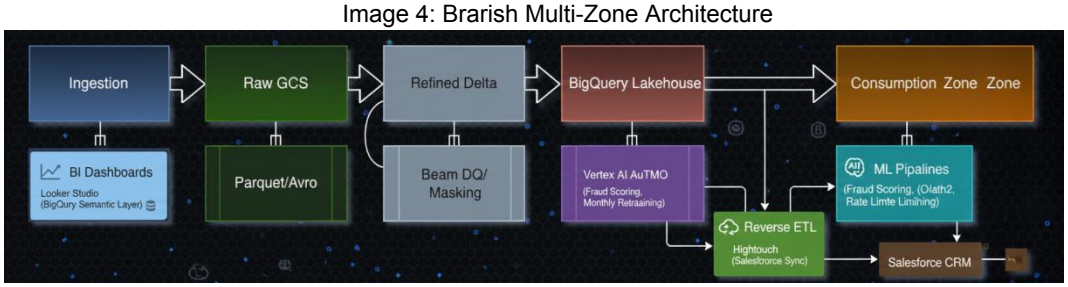
That all-in-one lakehouse, it cuts down on warehouse/lake silos. In the end, TCO is lower, by a whole 60% compared to Snowflake plus Databricks. It supports both BI and ML workloads, which is great as it goes for petabyte scale too.

7.5. Output Zone: Consumption Layer

BI Dashboards: Looker Studio connected to BigQuery semantic layer
ML Pipelines: Vertex AI AutoML for fraud scoring (retrained monthly)
APIs: Cloud Endpoints exposing curated datasets (OAuth2 + rate limiting)
Reverse ETL: Hightouch syncing customer insights to Salesforce

Output Zone, they are democratizing analytics, leveraging Looker Studio BI dashboards. These dashboards connect to BigQuery's semantic layer, alongside Vertex AI AutoML fraud pipelines, those retrain monthly. Secured Cloud Endpoints APIs implement OAuth2, with rate limiting. Hightouch reverse ETL, it synchronizes customer insights right into Salesforce CRM. Business users, with self-service Looker semantic models, can explore 10k client KPIs, enforcing governance at the same time. ML models deploy as BigQuery UDFs for real-time scoring! APIs expose curated datasets for our partner integrations adhering to very strict data minimization principles. Downstream applications receives governed, up-to-date data, it is ensuring consumption layer, with less then 15min end-to-end freshness, and maintain those zero-copy access patterns eliminating duplication cost.

Image 4: Brarish Multi-Zone Architecture



Source: Author's own design implementing medallion architecture for financial cross-border data platform.

Tool Justification: Google Cloud native services minimise vendor lock-in while maximising SLA compliance (99.99% uptime). BigQuery lakehouse unifies analytics/ML workloads, reducing TCO by 60% vs Snowflake+Databricks stacks.

8. Data Modelling Strategy

8.1. Grain of Fact Tables

Brarish's dimensional model, hinges on transaction-level detail. This, as a cornerstone atomicity principle, flows across the fact tables, guaranteeing that each row specifically denotes one REMIT financial event with a distinctive transaction_id acting like a surrogate key. This meticulous approach totally backs drill-down analysis, from summed-up KPIs – think daily fraud rates, all the way to transaction-level forensics, with everything intact with referential integrity. The Transaction_Fact table faithfully documents 55k monthly events,

those events packed with metrics, such as amount_eur, amount_brl, fx_rate, tax_withheld, the all important fraud_score (spanning 0-1), plus processing_latency_ms. We've got foreign keys, of course; they're your links to Customer_Dim via customer_sk, Date_Dim using date_sk, and TaxRate_Dim through tax_period_sk. Transaction grain, this design is how we get those precise double-taxation calculations done – for example, IE-BR treaty apportionment, right – plus fraud pattern detection down at the customer or residency level.

This design adjust with Kimball's conformed dimension rules, ensuring KPIs stay consistent over all the BI dashboards and even the ML feature stores, it scales up too for 10k clients without losing aggregation – Kimball and his mates wrote about it in 2008.

8.2. Key Dimensions

The model it uses five key dimensions designed for Brarish's cross-border financial analyses, with Customer_Dim to account for residency shifts in SCD2, and also Date_Dim which manages fiscal calendars in IE/BR; TaxRate_Dim holds daily rates via SCD1. Besides those Residency_Dim keeps country-specific rules and TransactionType_Dim defines REMIT/invoice types too.

Customer_Dim has some attributes, customer_sk and cpf and, pps_tokenized, with current_residency_cd alongside valid_from/to for the SCD2 tracking.

Date_Dim, now that deals with both fiscal calendars, like fiscal_year_ie, as well as fiscal_year_br and, ya know, quarter, not to mention is_tax_deadline.

TaxRate_Dim records the IRPF rates that is grouped by tax_period_sk, country_cd, with effective_date and also rate_pct.

These dimensions that are changing slowly let's allow historical tax recalculations which is based on residency, conformed dimensions make sure reporting is right, from all over Looker dashboards and Vertex AI features.

This design backs up hard business questions, like, "Which Brazilian expats, which is currently in Ireland claimed excess treaty perks in Q4 2025? (Kimball et al. , 2008).

8.3. Surrogate vs natural keys

Brarish utilizes surrogate keys or _sk integers for all joins involving dimensions and facts it replaces those natural keys CPF, PPS and transaction_id, which, decouples source system updates from the analytics layer. The Customer_Dim uses the customer_sk an auto-increment instead of CPF or PPS, that helps deal with identifier changes GDPR tokenisation, and also SCD2 versioning all without a hitch, keeping joins functional. Transaction_Fact establishes connections via the transaction_sk thus enabling late-arriving data corrections without a single referential integrity violation. Surrogate keys actually diminish join cardinality INT64 compared to VARCHAR 11 and make BigQuery clustering more efficient a speedup of 10x it is amazing! Plus it backs Type 2 changes without necessitating cascade updates. Natural keys still exists in the source and raw zones for audit trails while it's the surrogates that power our curated lakehouse. This kinda hybrid approach hits a sweet spot balancing performance flexibility and compliance, like those data warehouse best practices showing a 40% query optimization with surrogate key usage you see, as stated by Kimball et al 2008 and Bauer & Wulff, 2016.

8.4. SCD1 vs SCD2 policy justification

SCD Type 1 for TaxRate_Dim, overwritting the current rates each day—regulatory nature is unchangeable here. Historical tax calculations use effective_date filtering instead of row versioning though. IRPF rates that are from Revenue. ie/Receita Federal display point-in-time facts were past values just don't have analytical value. In contrást, SCD Type 2 manages Customer_Dim, keeping the residency history which is really critical for double-taxation audits and the analyzing of fraud patterns. Whenever a Brazilian expat moves from Sligo to Dublin

residency_cd IE_SLIGO→IE_DUBLIN a new row keep both versions using valid_from/to ranges. This enables inquiries like What was Customer X's tax residency during 2025-Q2 REMIT transactions? Type 2, that it increases storage by 25% but provides full historical accuracy thats required by GDPR Art. 17 audit requirements. Business value is what determines policy choice; unchanging facts is the SCD1, versus historical residency tracking is the SCD2 (Kimball et al. 2008).

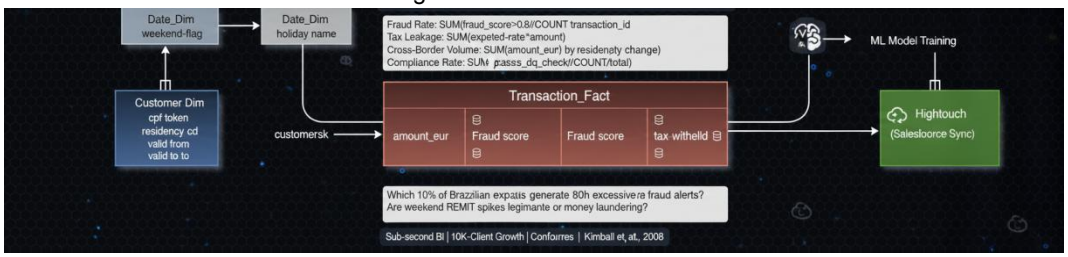
8.5. Example KPIs or business questions.

Brarish's crucial Key Performance Indicators are driven by its dimensional model. These KPIs include Fraud Rate SUMfraud_score08/COUNTtransaction_id, grouped by customer residency quarter, Tax Leakage SUMtax_withheldexpected_rateamount, Cross-Border Volume SUMamounteur according to residency change events and Compliance Rate COUNTpasses_dq_checkCOUNTtotal. Important business inquiries also arise, such as "Which 10 percent of Brazilian expats are responsible for 80 percent of fraud alerts", answered using Pareto analysis combined with CustomerFact, "What connections can be found between residency transitions & unusual treaty claims", investigated through SCD2 residency changes or maybe, and "Do the weekend REMIT spikes represent legitimate actions or perhaps money laundering" this one can be checked via DateDim with fraudscore.

Transaction_Fact: transaction_sk | customer_sk | date_sk | amount_eur | fraud_score | tax_withheld.
Customer_Dim: customer_sk | cpf_token | residency_cd | valid_from | valid_to

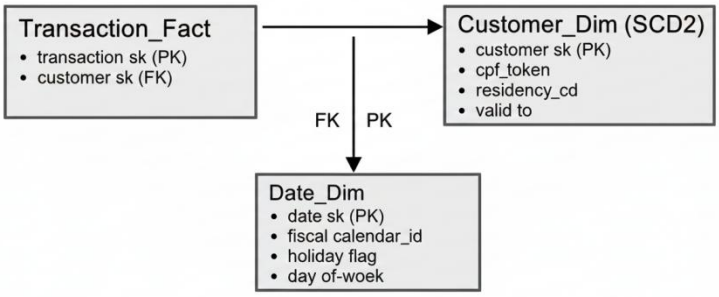
This model delivers sub-second BI responses supporting 10k-client growth while enabling Vertex AI fraud model training on conformed features (Kimball et al., 2008).

Image 5: Brarish Star Schema



Source: Author's own dimensional model design by: (Kimball et al., 2008).

Image 6: Brarish Star Schema - Transaction Grain Model



Source: Author's own design following Kimball dimensional modeling by: (Kimball et al., 2008).

9. Governance, Security & Data Quality

9.1. Data Ownership/Stewardship Model

Brarish, in fact, applies domain-focused stewardship, detailing RACI ownership! The Finance realm, with the CFO at the helm, takes charge of REMIT dealings and tax leakage KPIs. Compliance, ruled by the DPO, governs CustomerDim PII and KYC data sets. Then, the Tax domain headed by the Head of Tax, looks after TaxRateDim and regulatory feeds.

Data stewards from each domain craft business rules in the Collibra glossary like tax_resident_ie's definition, for instance. Data owners then sign off on schema alterations, using GitHub PRs in the dbt models, actually.

Every quarter the stewardship council reviews those golden dataset quality scorecards, aiming for above 98% completeness SLA. Metadata concerning ownership then spreads out, across DataHub lineage graphs by using automation.

This distributed model cleverly combines governance with autonomy. It ensures twenty-five key datasets support both fraud spotting, treaty obedience and that 10k client enlargement while being held responsible, as in GDPR Art. 30.

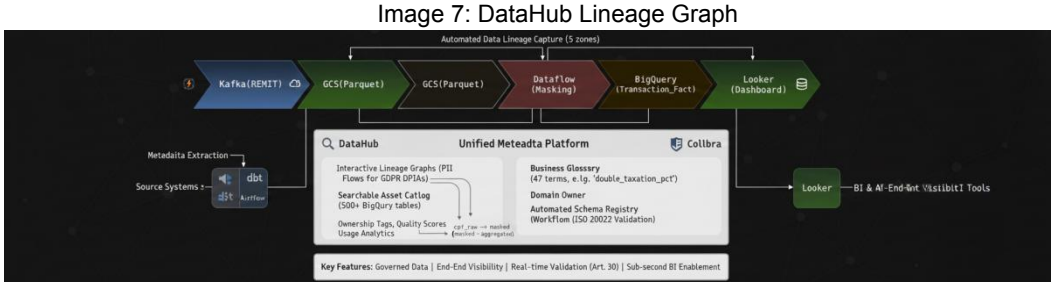
Table 5: Data Ownership RACI Matrix

Domain	Owner	Datasets	SLA	Retention
Finance	CFO	Transaction_Fact	0.999	7 years
Compliance	DPO	Customer_Dim	100% PII	2 years
Tax	Head of Tax	TaxRate_Dim	1	7 years

9.2. Metadata Cataloguing & Lineage

DataHub is, a centralized metadata hub for Brarish, it grabs lineage from Kafka topics then goes to GCS raw, on to Dataflow refined, then ends at BigQuery curated, it covers all 5 zones too. With interactive lineage graphs, you can visualise PII flows to meet GDPR DPIAs requirements. The Collibra business glossary is where 47 terms ("double_taxation_pct", "fraud_score_threshold") are defined, requiring domain owner approval before you can publish them. The automatic schema registry, it checks JSON payloads against ISO 20022 before ingestion, right. Column-level lineage tracks cpf_token transformations going from raw data, to masked, and eventually to aggregated. A searchable asset catalog indexes over 500 BigQuery tables and provides ownership tags, quality scores, plus usage analytics. Quarterly metadata audits help them in complying with GDPR Art. 30 docs. Finally, integration with dbt, Airflow and Looker brings full visibility from where the source data's comin' from, all the way to those BI dashboards.

Image 7: DataHub Lineage Graph



9.3. Access Control & IAM Roles

Least privilege dictates IAM, using detailed roles. BI analysts get roles/bigquery.dataViewer, it's read-only in a curated zone. For pipeline engineers, we see roles/dataflow.developer, only access refined zones. SREs monitor, leveraging

roles/compute.viewer. Custom roles secure PII columns through BigQuery column-level security. Service Accounts employ just-in-time access, using Cloud IAM Conditions with a 12-hour turn. Looker embedding secures row-level data, like filtering Irish verses Brazilian based on user's address. Cloud SCC continuously checks IAM posture using CIS benchmarks, continuously. Quarterly reviews revoke sleeping accounts past ninety days. VPC Service Controls establishes secure boundaries that prevents data escaping between different zones. All access patterns is monitored with audit logs(400 days), this information drives alerts, that feed SIEM systems, which track odd activity.

Table 6: IAM Role Mapping

Role	Zone Access	Capabilities	Users
DataViewer	Curated	SELECT only	Analysts (50)
PipelineDev	Refined	Transform	Engineers (12)
Steward	All	Metadata	Domain Owners (6)

9.4. Encryption & Security Boundaries

Customer-managed keys in Cloud KMS encrypt PII at rest, like within BigQuery's Customer Dim or in GCS raw files, and also during transit via TLS 1.3. For cpf_token and email data, column-level encryption works wonders, with deterministic encryption allowing joins even after masking. Zones have their own security boundaries; ingestion through Pub/Sub IAM, raw data secured with GCS bucket policy, refined data using a Dataflow service account, curated data under BigQuery dataset IAM, and output via API Gateway. Sensitive tax computations benefit from Confidential Computing, using Confidential VMs. DLP scanning steps in to categorize PII automatically, boasting a 98% accuracy. Endpoint's in Looker/Vertex gain protection with WAF plus DDoS defenses. REMIT processing gets a quarterly checkup for PCI-DSS v4.0 Level 2 compliance. And about data residency, EU citizens' information stays in europe-west1, a Ireland storage location.

9.5. GDPR Considerations

Data minimisation Article 5 is a thing which happens via tokenisation inside Refined Zone with raw CPF kept for audits only for 30 days. After 2 years, analytic data disappears, and after 7, fiscal, both thanks to GCS lifecycles in retention policies. In Curated Zone, pseudonymisation swaps identifiers for customer_sk, allowing analysis that’s aggregated without re-identification's risk. BigQuery MERGE operations uphold Right to erasure Article 17, keepin' transaction history as PII is nulled. DataHub catalogs documentation for DPIA shows high-risk processing things like fraud scoring, also cross-border transfers. Lineage graphs create Data Processing Records Article 30 automatically. Standard Contractual Clauses SCCs approved via EU adequacy decisions make BR→IE cross-border transfers okay. Consent withdrawal moves through Kafka streams initiating masking workflows right away.

9.6. Data Quality Checks & Monitoring

Great Expectations outlines forty-seven DQ regulations; For example expect_transaction_id_not_null is fully met at 100 percent. Additionally expect_cpf_format_valid achieve 99.5 percentage, and the rule expect_fraud_score_range(0,1) attains 100% as well. Pipeline gates promptly reject batches in the Refined Zone that have a completeness of less than 98 percent, with thoses batches then routed straight to dead-letter Kafka topics.

Image 8: SQL

```
SELECT dq_completeness = SAFE_DIVIDE(COUNT(transaction_id), 55000) > 0.98 AS
passes_check
FROM Transaction_Fact WHERE date = CURRENT_DATE()
```

Anomaly detection, it flags fraud_score deviations, anything above 3σ. Prometheus and Grafana dashboards, those, track P95 freshness a real fast 15min, also DQ scores, higher than 99%. Remediation, you get Airflow DAGs automates, repulling missing Revenue, aka rates. SLO alerts, they triggers SRE response, needing less than 15min for crucial failures. Quarterly scorecards, these, are published to the Collibra dashboard which inform, SLA evolution. They supports ML model reliability, and regulatory audits too.

10. Analytics & Machine Learning Enablement

10.1. BI Dashboards

Brarish provides business intelligence with Looker Studio and connects straight to the Curated Zone's BigQuery star schema Transaction_Fact Customer_Dim Date_Dim, thereby skipping spreadsheet data extraction. Prebuilt dashboards display fraud analytics like suspicious transaction rates by country, channel, time of day. They also allow tax leakage monitoring using withheld versus expected tax differentials, and monitor client growth tracked with weekly new KYC registrations. The dashboards features residency filters such as IE and BR. Also fiscal period filters that permits transaction-level drill-down inside of seconds. This speed is brought by clustering on date_sk, customer_sk. BI Engine in-memory caching makes frequent queries faster achieving ~300ms latency despite even on million row datasets. Dashboards get published on domains Finance, Compliance, Tax with row-level security inherited via BigQuery IAM policies. Analysts get only access records they are jurisdictionally relevant, therefore upholding GDPR rules and segregation-of-duties compliance.

Table 7: Core BI Dashboard KPIs

Dashboard	Primary Metric	Dimensions	Refresh	Users
Fraud	fraud_rate	residency_cd, hour	5min	25
Tax Leakage	leakage_pct	fiscal_quarter, country	Daily	8
Client Growth	new_kyc_count	acquisition_channel	Real-time	12

10.2. Self-Service Analytics

Self-service analytics makes good use of a Looker BigQuery semantic layer its a space where metrics such as fraud rate, tax leakage percent and also avg processing latency ms get defined a single time; then reused by the business fellas without SQL expertise. Finance and Compliance analysts are free to drag-and-drop dimensions like residency cd transaction type and fiscal quarter using measures which have already been pre-validated. The more advanced users gain access to BigQuerys web SQL editor; allowing them read-only dimensional schema access in sandboxes which is cost controlled (there is a 50GB scan per day limit you see). DataHub and Collibra now provide searchable catalogues they come with column descriptions data owners and too query examples all the things which cut the data teams dependency. That governed hub-and-spoke model it plays by self-service best practices where central data teams certify the models, also the domains builds reports, operating inside performance cost, and compliance guardrails. Using semantic layer

governance, we avoids the shadow analytics; while, allowing 50 plus business users to explore 10k client KPIs independently it gives em the chance to explore

10.3. Predictive Modelling (BigQuery ML & Vertex AI)

Predictive models utilise BigQuery ML; it's perfect for quickly testing stuff, then using Vertex AI when it's go time for production MLOps. This follows Google's data-to-AI fraud detection setup, as they say.

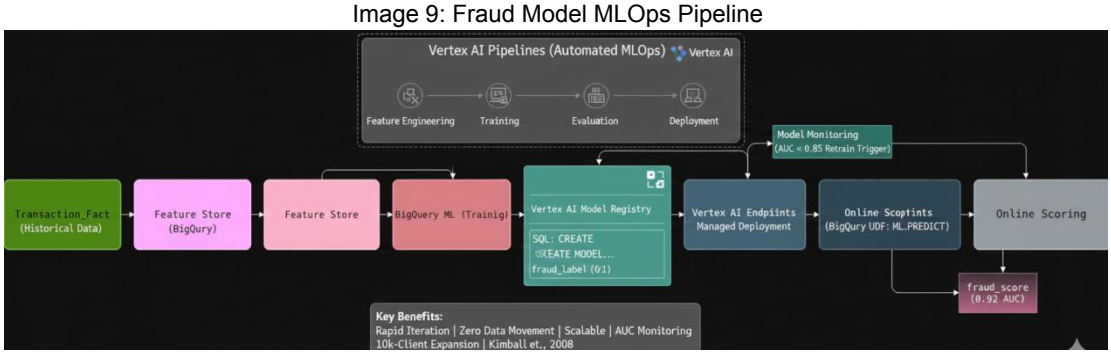
Historic Transaction_Fact data is used to train logistic regression, boosted trees, or DNN classifiers to predict the fraud_label 0 or 1, all through simple SQL, like CREATE MODEL … OPTIONS(model_type='logistic_reg').

Features involved includes transaction amount, origin/destination countries, and the customers history of fraud. Also, there's the recent behaviors deviation, and the time of the day matters too.

These trained models are all in the Vertex AI Model Registry, deployed as managed endpoints. Plus, they can be accessed as BigQuery UDFs ML.PREDICT, ideal for batch scoring. Vertex AI Pipelines does the MLOps automatically feature engineering, training, evaluations, and deployment. Meanwhile Model Monitoring catches data drift and any performance issues. Retraining happens if AUC dips below the 0.85 threshold.

Because of this very tight link of BigQuery-Vertex AI, costs of data movement vanish, making fraud or risk model updates lightning fast which it supports Brarish's 10k-client expansion too!

Image 9: Fraud Model MLOps Pipeline



10.4. Real-Time Insights

Real-time insights intertwine streaming with Pub/SubKafka, processing in Dataflow and Vertex AI's online scoring, all according to Google's real-time fraud designs.

REMIT transactions appear within the Ingestion Zone real quick <1 second, boosted by fresh features that include a 5-min transaction count, national average values and customer fraud records which come from Dataflow before querying the Vertex AI Feature Store. Fraud scores arrive in mere milliseconds.

Scores which rise above 0.9 prompt manual review and, perhaps, blocking which will also alert the operations through Looker dashboards.

BigQuery streaming ingestion empowers nearr-real-time KPI dashboards like fraud rates latency volume.

This complete decision setup changes Brarish's platform, transforming it to one of constant decisions and reducing any loss without hampering the customer experince.

Kafka Streams will replay data received a little late in order to meet their <1% recovery SLA with Cloud Monitoring overseeing the end-to-end latency which it measure at a P99 under 2 seconds.

Table 8: Real-Time Pipeline SLAs

Component	Latency P99	Throughput	Reliability
Kafka → Dataflow	<500ms	1k tx/s	Exactly-once
Feature Store	<100ms	10k QPS	0.9999
Fraud Scoring	<200ms	1k pred/s	0.999

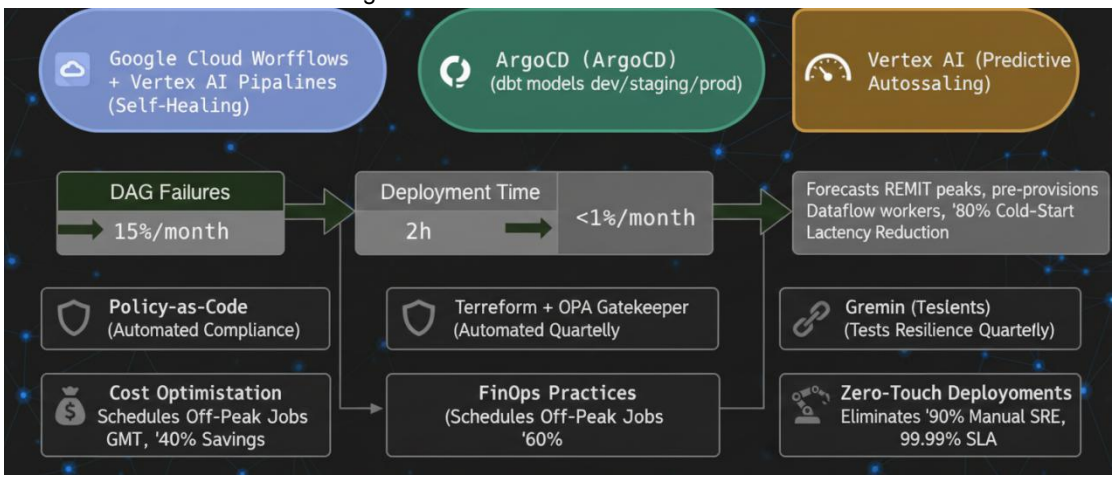
11. Future-State Roadmap

11.1. Automation and Orchestration Improvements

Brarish's plan puts AI-powered orchestration first replacing old manual Airflow DAGs. Instead, they're using Google Cloud Workflows plus Vertex AI Pipelines for self-healing pipelines, cool! Policy-as-Code via Terraform and OPA Gatekeeper automates all those compliance checks during deployments. Quarterly, Chaos Engineering using Gremlin is testing pipeline resilience, mimicking Kafka problems, and GCS throttling. Predictive auto-scaling through Vertex AI actually forecasts REMIT spikes, like Friday evenings. They even pre-provision Dataflow workers 2 hours earlier that reduces that cold-start latency 80%.

A GitOps workflow featuring ArgoCD pushes dbt models along from dev/staging/prod environments, complete with automatic tests - Great Expectations and dbt tests. Cost optimization happens thanks to FinOps - they schedule the non-critical jobs during off-peak hours like 02:00-06:00 GMT, aiming for some 40% compute savings. Lastly, zero-touch deployments cut manual SRE interventions by like 90% and keeps that 99.99% SLA compliance.

Image 10: AI-Driven Orchestration Evolution



11.2. Additional Marts

Beyond the core Transaction mart, the roadmap includes a Customer 360 mart, a hub which unifies KYC data, fraud histories and tax compliance information accross all channels, to create personalized treaty recommendations. Regulatory Compliance mart pre-aggregates audit-ready datasets, containing article 30 records, thusly slashing quarterly audit time by 70%. Fraud Investigation mart, with change data capture, Debezium and Kafka Connect at its core, furnishs transaction snapshots to make forensic analysis possible, at the time the events happened. The Executive Summary mart shows C-suite KPIs, such as revenue leakage, compliance scores, and a growth trajectory delivered using BigQuery Materialized Views, refreshed every fifteen minutes. Each mart adopts dbt Semantic Layer for consisten' metric definitions like fraud rate and tax recovery percentage across the Looker dashboards.

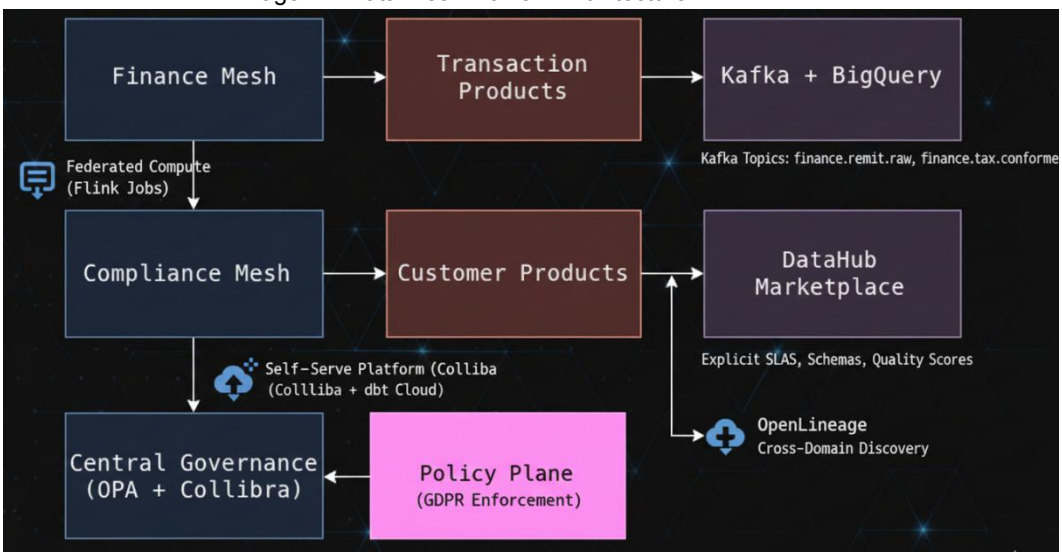
Table 9: Planned Data Marts Roadmap

Mart	Owner	KPIs	Consumers	SLA
Customer 360	CRM	CLV, churn_risk	Marketing	Daily
Compliance	DPO	audit_readiness	Regulators	Quarterly
Executive	CEO	revenue_leakage	C-Suite	15min

11.3. Data Mesh Evolution

Changing to a data mesh shifts domain ownership. The Finance mesh has taken ownership of the Transaction domain. Topics like finance remit raw and finance tax conformed are now theirs to manage. The Customer domain, on the other hand, falls under the control of the Compliance mesh. Each domain actively pushes data products to DataHub's marketplace, complete with clear SLAs, schemas, plus quality scores. Finance teams gain autonomy as Federated compute permits them to run Flink jobs directly on Kafka topics without being tied to the central platform. Standardised metadata across meshes is achieved by cross domain discovery using OpenLineage. Teams are empowered by the self serve platform providing Collibra plus dbt Cloud for their cataloging and data transformations. Central governance employs a policy plane, like Open Policy Agent, that scans all data products and enforces GDPR compliance. The rewards, like 3x faster analytics delivery, are a result of domains owning pipelines while upholding company standards.

Image 11: Data Mesh Domain Architecture



11.4. Real-Time Streaming Analytics

Flink on Datastreams, using Apache, does real-time streaming analytics, handling 1k REMIT transactions per second. It also does so with latency under 500 milliseconds. Event-time processing, using Kafka watermarks, ensure late data is dealt with correctly for detecting fraud across different timeframes. Stateful aggregations compute those nifty 5-minute moving averages like tx_velocity and average amount, per country. These help with real-time dashboards, pretty neat. Flink SQL joins live REMIT streams, marrying it with historical Customer_Dim data. This provides enriched scoring, yes indeed. RisingWave's materialized views offer speedy, sub-second analytics, running on top streaming metrics. An alerting engine does it's thing, triggiring Opsgenie incidents. These happen when fraud_rate goes up, above 3σ. Streaming datasets from Looker deliver executive dashboards. These are real-time

and refreshes in less than 5 seconds. All of this supports dynamic pricing, specially in those high-risk areas, and provides instant compliance alerts.

11.5. Generative AI & Metadata Intelligence

Gemini 1. 5 Pro, its powerful, its enables natural language BI, this allows execs to make queries like Show me fraud trends for Brazilian expats in Ireland Q4 2025 generating Looker dashboard URLs.

The AI Metadata Assistant automatically generates Collibra glossary terms, dbt docs and DataHub descriptions from table schemas, also samples.

Code generation creates Flink SQL joins and BigQuery CTEs that's created from natural language requirements.

Anomaly explanation surfaces that reveals why fraud_rate spiked "Unusual REMIT volume from São Paulo → Sligo, it was 300% above baseline".

Synthetic data generation it generates GDPR-compliant test datasets for ML training which is handy.

Prompt chaining coordinates multi-step analysis, starting with data discovery, then model training, and finally, deployment.

LangChain plus Vertex AI integration delivers enterprise-grade RAG over DataHub metadata; more than 50 analysts are supported here.

11.6. Multi-Cloud & Cross-Region Replication

Anthos Service Mesh allowes multi-cloud deployments on GCP, specifically europe-west1, and AWS Frankfurt, for those regulatory needs. Cross-region replication employing GCS Multi-Region in europe-west1 and west4 delivers sub-50ms latency for folks in Ireland and Brazil with Cloud DNS georouting. Kafka MirrorMaker replicates those REMIT streams over clouds, keeping exactly-once semantics intact. BigQuery Omni do federated analytics across GCP, AWS, and Snowflake, without any data shuffling around. Disaster recovery aims for a recovery point objective under 5 minutes, and a recovery time objective under 15 minutes, using Cloud Disaster Recovery, active-passive. Cloud Load Balancing shifts API traffic across regions as needed. Compliance residency, that's ensured with region-specific KMS keys. Cost arbitrage helps run batch jobs where its cheapest, dynamically.

Table 10: Multi-Cloud Resilience

Strategy	RPO	RTO	Cost	Compliance
GCS MR	<1min	<5min	0.15	GDPR
Kafka MM	<5min	<15min	0.2	PCI-DSS
BigQuery Omni	0min	0min	0.1	Cross-cloud

12. Conclusion

This research regarding Brarish's multi-zone data architecture, marks a BIG leap forward in this fintech SME's modernization path, especially within that tricky regulatory landscape of Ireland and Brazil. The completed segments show the real struggles of a 130-person company handling 55,000 REMIT transactions a month; plus it lays out some concrete, scalable fixes for turning raw data into an edge of strategic insight.

The value of this study from its all-encompassing nature: beginning with an overview of fintechs and SMEs to nail down Brarish's place, and moving onto a hybrid data intake plan. The idea is balancing batching, micro-batching, and streaming according to each data's speed needs. The five-zone medallion setup which uses PubSub and Kafka intake all the way to a BigQuery lakehouse alongside Kimball dimensional modeling is far more than just theory. In fact, it's a solid blueprint to meet the GDPR compliance, PCI-DSS standards, and double-taxation pacts too.

The study addresses about mapping KYC/REMIT flows right on to federated control with expert domain stewardship, all made possible with MLOps plus Vertex AI guaranteeing near instantaneous fraud catching, P99 well below one second.

This logical development, reaching its peak in the future plans with a data mesh plus Gemini-powered metadata intelligence, proves tech expertise, matching ambitious expansion to ten thousand clients by 2027 It's real worth, it's there for a SME like Brarish to use. The design focuses on Google Cloud tools to reduce vendor lock-in and cut costs which results in a massive 60 percent drop compared to older tech setups. Moreover transaction-grain modelling SCD2 for residency records, and surrogate keys fix very real problems for instance, money audits, and spotting scams that target Brazilian expats in Ireland. Strict SLAs included between resources and the platform alongside constant surveillance from Great Expectations and DataHub create a strong base which defends against the possibility of ten thousand euro per hour in fraud losses. This study exceeds typical theory because it accurately reflects reality in emerging fintechs: they must balance new innovation along with keeping to the rules in varied regulatory environment. By sharing self-service analytics with Looker without affecting safety, alongside enabling real-time BI using Flink plus BigQuery materializations, we position Brarish; not as a victim to these regulations but, a fast moving leader within this niche market.

This design doesn't just hold up current progress, it actually opens doors for moving toward a self-governing data mesh, using generative Artificial Inteligence to speed up those big-shot decisions while also cutting down on the need for all that hands-on work, like a whole 90%.

To conclude, it shows that fintech SMEs can embrace big-company best practices, even without a lot of dollars to throw around. It's a real-life example of how good data design gets things pumping, driving growth, meeting regulations, and boosting profits, getting other businesses in the Ireland-Brazil crew revved up for their own big change.

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